

# Final Project Document

Tech Stack: **Flutter + Firebase + Provider**

Restrictions:

- No file uploads
  - No push notifications
  - No platform-specific features
  - Uses only Firebase Firestore (no storage)
  - Beginner-friendly CRUD + simple UI
  - Phone-only, no backend server
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## Project List

| Project ID | Name                                            |
|------------|-------------------------------------------------|
| P004       | E-commerce Mini Shopping Cart (Unsplash Images) |

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## P004 — E-commerce Mini Shopping Cart (Unsplash Images)

### Project Story

A tiny online store app with product browsing and cart management. All product images come from preset Unsplash URLs (no file upload).

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### Functionalities

1. Product Catalog Load from Firebase Each product has name, price, category, image URL
2. Product Details Larger image Name, price, description Add to cart
3. Cart Screen Increase/decrease quantity Remove item Auto-update totals
4. Order Placement Place order with customer details (name, phone, address)
5. Order confirmation screen Order with status (Order placed)

6. Order ID generation (Firebase Document ID)
  7. Order History View all placed orders Filter orders by status View order details
  8. Checkout Save order summary in Firebase Clear cart when done
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## Firestore Structure

```
None
/products
  /productId
    name
    price
    category
    imageUrl

/orders
  /orderId
    items
    total
    date
    status
    customerName
    customerPhone
    customerAddress
    orderId
```

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## P005 — Simple Expense Calculator

### Project Story

A lightweight expense tracker app. Users can add expenses, view totals, filter by category, and manage their expense entries.

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### Functionalities

1. Add Expense Name Amount Date Category Description (optional)
2. Expense List all entries Swipe to delete Edit expense entries
3. Total Expense Auto-calculated total from all expenses Category-wise total display

4. Category Filter expenses by category Default categories: Food, Transport, Entertainment, Shopping, Bills, Other
  5. Search Expenses by name
  6. Sort Expenses Sort by date (newest first) Sort by amount (high to low)
  7. Firebase Sync All expenses stored in Firestore Real-time sync across devices
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## Firestore Structure

None

```
/expenses
  /expenseId
    name
    amount
    date
    category
    description
    createdAt
```

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